

In Excel, managing large datasets—such as those with up to **1,048,576 rows**—requires tools that can handle filtered data and hidden rows without returning errors. Both **SUBTOTAL** and **AGGREGATE** are designed to perform calculations on a range while ignoring rows that are not visible.

1. The SUBTOTAL Function

The **SUBTOTAL** function is the traditional method for summarizing filtered lists. It uses a "function_num" code to determine which operation to perform (e.g., Sum, Average, Count).

- **Syntax:** `=SUBTOTAL(function_num, ref1, [ref2], ...)`
- **Behavior Codes:**
 - **1-11:** Includes manually hidden rows but ignores rows hidden by a filter.
 - **101-111:** Ignores **all** hidden rows (both manually hidden and filtered out).
- **Common Codes:**
 - **9 or 109:** SUM
 - **1 or 101:** AVERAGE
 - **2 or 102:** COUNT
- **Limitation:** **SUBTOTAL** cannot ignore error values like `#DIV/0!` or `#N/A`. If your range contains an error, the formula will also return an error.

2. The AGGREGATE Function

Introduced as a more powerful version of **SUBTOTAL**, the **AGGREGATE** function can handle errors and offers a wider variety of operations.

- **Syntax:** `=AGGREGATE(function_num, options, ref1, [ref2], ...)`
- **The "Options" Argument:** This is the "secret sauce" of **AGGREGATE** that allows you to specify what to ignore:
 - **0:** Ignore nested **SUBTOTAL** and **AGGREGATE** functions.
 - **3:** Ignore hidden rows, error values, and nested functions.
 - **4:** Ignore nothing.
 - **6:** Ignore only error values.
- **Expanded Operations:** In addition to basic sums and counts, **AGGREGATE** can perform 19 different functions, including **MEDIAN**, **PERCENTILE**, and **LARGE**.

3. Comparison: When to Use Which?

Feature	SUBTOTAL	AGGREGATE

Ignore Filtered Rows	Yes	Yes
Ignore Manually Hidden Rows	Yes (using 100-series codes)	Yes (using specific options)
Ignore Error Values	No	Yes (Option 6 or 3)
Number of Functions	11 functions	19 functions
Nested Functions	Automatically ignores other SUBTOTAL functions in the range.	Can be set to ignore or include other AGGREGATE functions.

4. Practical Integration for Data Projects

- **Real-time Dashboards:** Use **AGGREGATE** with Option 3 when building a "Sales chart" or "Salary statement" so that your totals update instantly as users filter the data, even if some rows contain #N/A errors from a failed **Lookup function**.
- **Data Cleaning:** Use **SUBTOTAL** when you want a quick count of visible records after applying an **Advanced Filter** to a list of student records.
- **Formula Auditing:** Both functions help in **Formula auditing** by ensuring that "Double Counting" does not occur when you have sub-totals within a larger master list.